Ethics and Broader Impact

Data Collection While we generated the facts in our examples, the logical rules have been mined from the data in the DBpedia knowledge graph, which in turn has been generated from Wikipedia.

Biases We are aware of (*i*) the biases and abusive language patterns (Sheng et al., 2019; Zhang et al., 2020; Bender et al., 2021; Liang et al., 2021) that PLMs impose, and (*ii*) the imperfectness and the biases of our rules as data from Wikipedia has been used to mine the rules and compute their confidences (Janowicz et al., 2018; Demartini, 2019). However, our goal is to study PLM's capability of deductive soft reasoning. For (*i*), there has been some work on debiasing PLMs (Liang et al., 2020), while for (*ii*), we used mined rules to have more variety, but could resort to user-specified rules validated by consensus to relieve the bias.

Environmental Impact The use of large-scale Transformers requires a lot of computations and GPUs/TPUs for training, which contributes to global warming (Strubell et al., 2019; Schwartz et al., 2020). This is a smaller issue in our case, as we do not train such models from scratch; rather, we fine-tune them on relatively small datasets. Moreover, running on a CPU for inference, once the model is fine-tuned, is less problematic as CPUs have a much lower environmental impact.

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